

Learned performance or auditory bias: carryover vs. anticipatory nasal coarticulation

Ian Carpick, Emily Knick, Cal Boye-Lynn, Ryan Bennett, & Amanda Rysling

icarpick@ucsc.edu | UC Santa Cruz Linguistics | LabPhon 19, July 29 2024



Background and research questions

Nasal coarticulation: vowels (V) are (partly) nasalized adjacent to nasal consonants (N) [3].

- Carryover nasal coarticulation: /NV/ → [N[̃]V] with gradually decreasing nasality.
- Anticipatory nasal coarticulation: /VN/ → [V[̃]N] with gradually increasing nasality.

Asymmetric perception of nasal coarticulation?

Listeners can use both carryover and anticipatory nasal coarticulation in vowels as evidence during laboratory tasks [1, 2, 7].

But it has been claimed that listeners use anticipatory more often than carryover nasal coarticulation [6] — a **perceptual asymmetry** — despite the latter being more articulatorily [4] and cross-linguistically [8] extensive.

Research questions: In a head-to-head comparison, **is** there a perceptual asymmetry? If so, to what extent is it due to properties of general auditory perception versus the experience that listeners have of their most expert language(s)?

Experiment 1: categorization with gated stimuli (*n* = 10)

Q: Do native English listeners use carryover nasal coarticulation as evidence for previous nasal consonants as readily as they use anticipatory nasal coarticulation as evidence for upcoming nasal consonants?

A: No; with partial evidence, they make categorization decisions based on anticipatory but not carryover nasal coarticulation, showing that there is an asymmetry somewhere in the perception of nasal coarticulation.

Forced choice categorization paradigm with gated stimuli:

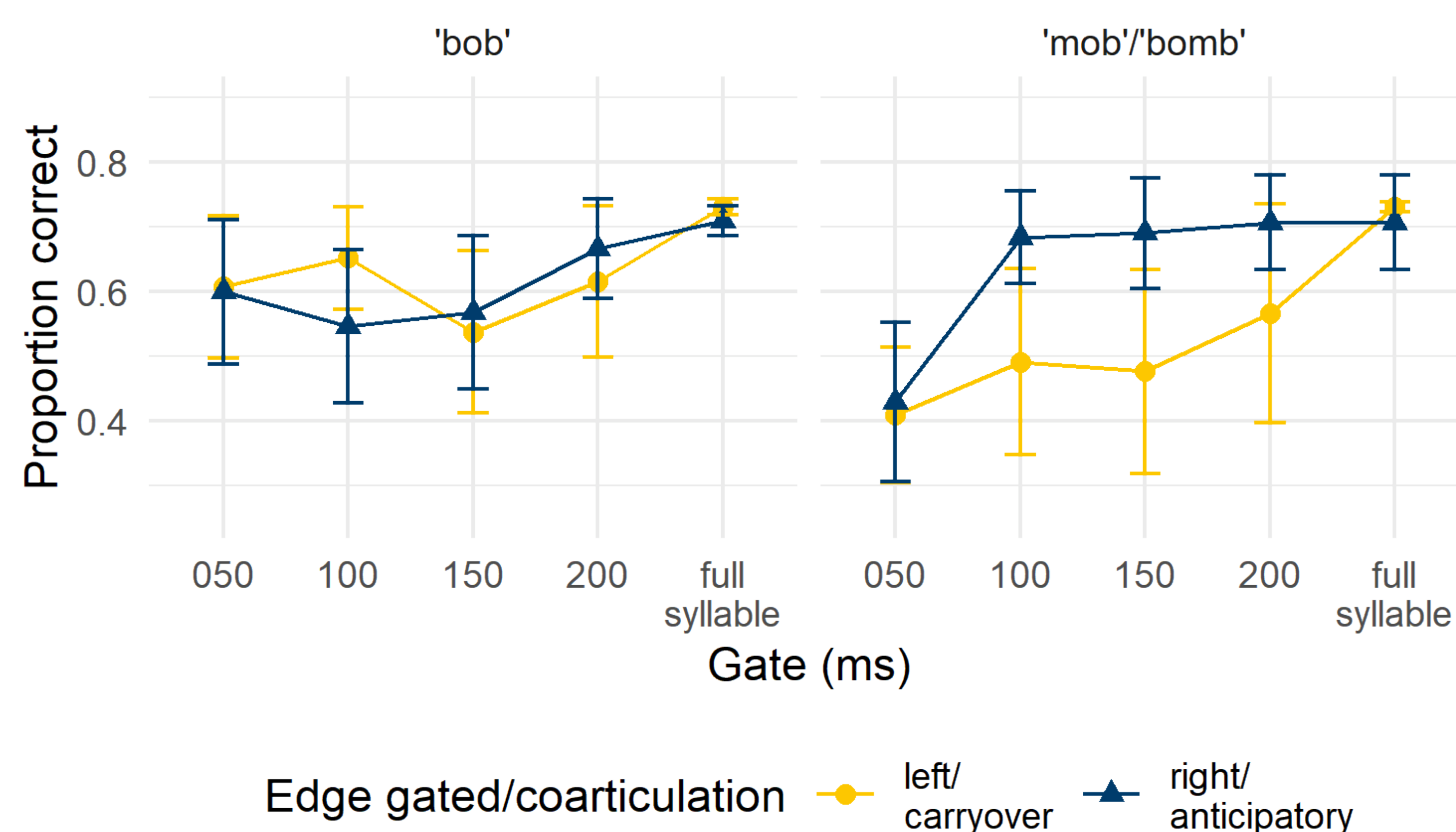
- Listeners heard recordings of ‘bob’, ‘mob’, ‘bomb’ gated into noise.
- Listeners categorized the stimuli as oral or nasal based on (the lack of) nasal coarticulation.

Factors: 2 × 2 × 5.

- **2 token types:** oral ‘bob’ versus nasal ‘mob’/‘bomb’ (depending on direction)
- **2 gating/coarticulation directions:** left-edge gating (carryover) ‘bob’ vs. ‘mob’, right edge gating (anticipatory) ‘bob’ vs. ‘bomb’
- **5 gate lengths:** 50 ms of vowel, 100 ms, 150 ms, 200 ms, full syllable

Results: Directly comparing the perception of carryover versus anticipatory nasalization revealed an asymmetry in native English listeners’ willingness to infer a nasal consonant.

- In the oral condition, performance wasn’t affected by gating direction...
- ... but in the nasal condition, performance was much better with anticipatory than carryover.



Edge gated/coarticulation — left/carryover — right/anticipatory

Two approaches to perceptual asymmetry

- Why did native English listeners categorize based on anticipatory but not carryover?
- Is this due to their English experience, or would other listener groups behave similarly?

Hypotheses and predictions

Auditory bias hypothesis: Carryover nasalization is harder to detect than anticipatory due to a general auditory effect (e.g., habituation [9]).

Prediction: listeners should perceive carryover nasalization less veridically than anticipatory, regardless of language-specific knowledge.

Language-specific knowledge hypothesis (not directly tested): Native English listeners prioritize evidence from anticipatory nasalization and deprioritize evidence from carryover, because evidence for upcoming but not previous nasals speeds word recognition.

Prediction: listeners utilize carryover versus anticipatory nasalization depending on whether they disambiguate words in the listeners’ language.

Experiment 2: 4IAX discrimination (*n* = 31)

Q: Do native English listeners perceive carryover nasalization less veridically than anticipatory, i.e., is there a general auditory effect inhibiting the detection of carryover?

A: No; in fact, they perceive carryover **more** veridically, the opposite of what is expected under the auditory bias hypothesis (though this result is not predicted by the language-specific knowledge hypothesis, either).

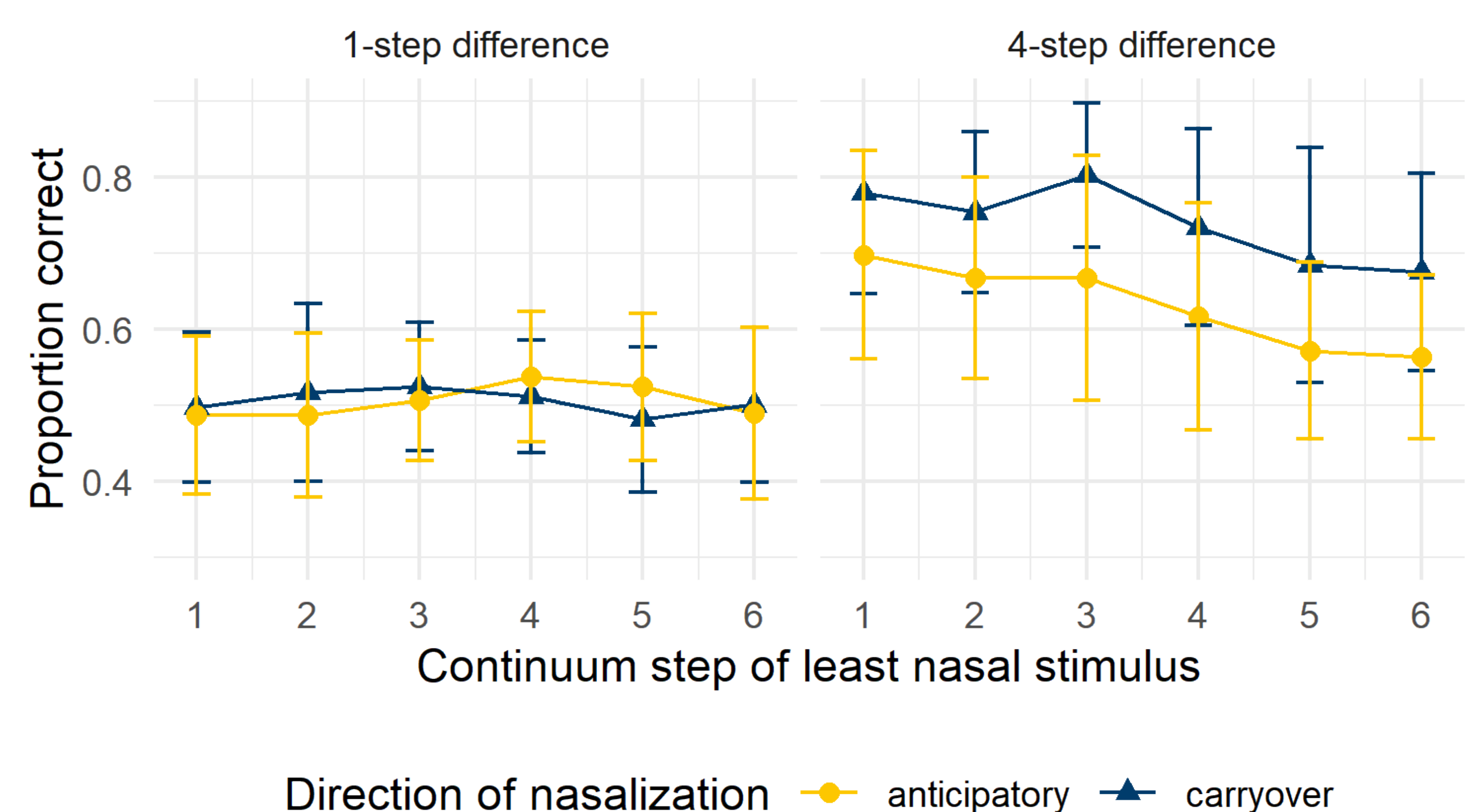
Forced choice 4-interval AX (4IAX) discrimination paradigm:

- Listeners heard two pairs of vowels (four total) from ten-step continua between fully oral and heavily nasalized (either carryover or anticipatory) in a [b_b] frame and reported which pair contained a difference.
- E.g., the correct response to ‘V₁ V₁ — V₂ V₁’ is ‘second pair is different.’
- The 4IAX paradigm biases listeners towards non-categorical perception [5].

Factors: 2 × 2 × 6.

- **2 coarticulation directions:** carryover, anticipatory
- **2 difference sizes:** 1-step difference, 4-step difference
- **6 base continuum steps:** steps 1 through 6, 1 being least nasal

Results: The primary factor affecting performance was the size of the difference between stimuli, not direction of nasal coarticulation.



Discussion

- Direct comparisons of the perception of carryover and anticipatory nasal coarticulation found evidence that...
- Native English listeners use anticipatory but not carryover nasal coarticulation as evidence for nasal consonants in a categorization task...
- ... but it's not harder for them to perceive carryover than anticipatory nasal coarticulation.
- Thus, the “perceptual” asymmetry is not (primarily) attributable to a general auditory effect that is active whenever listeners hear nasalization...
- ... and must instead be attributed to some application of listeners’ experience, though the nature of that knowledge is not clear.

Conclusions

- We find evidence for asymmetric use of carryover and anticipatory nasalization.
- We find evidence against an auditory bias hypothesis to explain this asymmetry.
- Some kind of experience-based explanation for asymmetric responding is indirectly supported.

Future directions: test specific models of the auditory bias hypothesis; directly test the language-specific knowledge hypothesis.

References

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